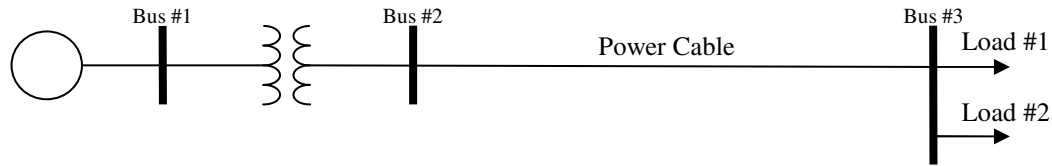
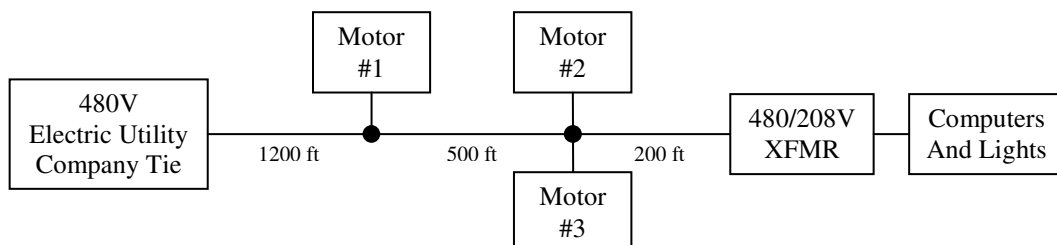


1. In the single phase power system shown in the one line diagram below, a source supplies power through a step-up transformer and then a 3750ft length of underground power to two parallel connected loads.



The source operates at 485V and supplies 40KVA at a 0.92 lagging power factor. The power cable has net distributed series impedance of $0.002552\angle 11.507^\circ \Omega/\text{ft}$. The step-up transformer is rated 50KVA, 480V/8KV, has 2.6% impedance, and has a 1.69 X/R ratio. Compute the following system quantities.

- Real and reactive power supplied by the source. (36.8000KW, 15.6767KVAR)
 - Voltage, current, and power factor on the high voltage side of the transformer. (7.9512KV, 4.948A, 0.924738 lagging)
 - Voltage, current, and power factor on the load end of the power cable. (7.9047KV, 4.948A, 0.924307 lagging)
 - Percent loading and efficiency of the transformer. (78.693%, 98.872%)
 - Efficiency of the power cable. (99.369%)
 - Power system efficiency. (98.248%)
-
2. For the system in Problem #1, assume the source is an electric utility company tie rated 480V with a short circuit current of 25kA, X/R ratio of 8, and 489.25V no-load operating voltage. Also assume that the impedances of Load #1 and Load #2 are respectively $1100\angle 60^\circ \Omega$ and $1500\angle -80^\circ \Omega$. Compute the following quantities for this system.
- Voltage, current, and power factor at the electric utility company tie point. (488.716V, 77.942A, 0.971552 lagging)
 - Real and reactive power supplied by the electric utility company to the system. (37.0079KW, 9.0211KVAR)
 - Voltage, current, and power factor on the high voltage side of the transformer. (8.0373KV, 4.677A, 0.974740 lagging)
 - Voltage, current, and power factor on the load end of the power cable. (7.9926KV, 4.677A, 0.974710 lagging)
 - Percent loading and efficiency of the transformer. (75.173 %, 98.998 %)
 - Efficiency of the power cable. (99.440%)
 - Power system efficiency. (98.444 %)
-
3. In the three phase power system shown in the block diagram below, a 480V electric utility company tie supplies power to a small manufacturing facility. In this facility, three Y-connected motors are powered from the nominal 480V level, and a step-down transformer is used to create a nominal 208V level for computers and lighting. A simple network of power cables is used in the 480V part of the system.

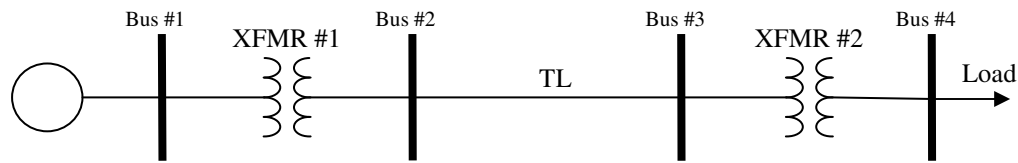


The 480V electric utility company tie has an equivalent source impedance of $0.02\angle 85^\circ \Omega$. The power cable used in this system has a distributed series impedance of $20\angle 39^\circ \mu\Omega/\text{ft}$. The per phase impedances of the

three motors are respectively $15\angle 75^\circ\Omega$, $16\angle 80^\circ\Omega$, and $19\angle 70^\circ\Omega$. The step-down transformer is rated 5KVA, 480V/208V, Δ -Y, $r_1=0.1\Omega$, $x_1=1.1\Omega$, $r_2=0.03\Omega$, $x_2=0.3\Omega$. The computer and lighting load is Y-connected with per phase impedance of $12\angle 30^\circ\Omega$. Compute the following system quantities.

- Voltage, current, and power factor at the electric utility company tie point. (478.212V, 52.970A, 0.310056 lagging)
- Real and reactive power supplied by the electric utility company to the system. (13.6036KW, 41.7125KVAR)
- Current in each of the three power cables. (52.970A, 34.671A, 4.222A)
- Voltage at each of the three motors. (476.366V, 475.854V, 475.854V)
- Voltage, current, and power factor on the high voltage side of the transformer. (475.825V, 4.222A, 0.853433 lagging)
- Voltage, current, and power factor on the low voltage side of the transformer. (202.486V, 9.742A, 0.866025 lagging)
- Percent loading and efficiency of the transformer. (68.335%, 99.652%)
- Power system efficiency. (98.563%)

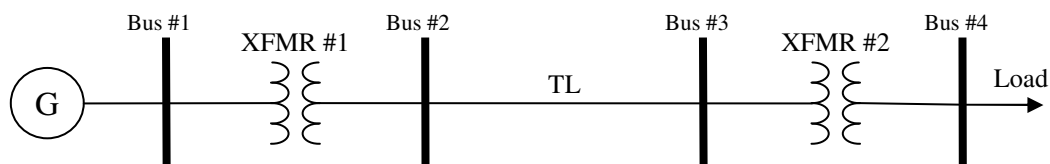
4. The one-line diagram for a single phase power system is shown below. In this system a source supplies power to a remotely located load through a high voltage transmission line. One transformer is used to step-up the source voltage to create a high voltage level for the transmission line, and another transformer is used to step-down the high transmission line voltage to the level required for the load.



The source operates at 18.25KV and supplies 2.55MVA at a 0.913752 lagging power factor. The transmission line has a total series impedance of $9.75\angle 55^\circ\Omega$. The step-up transformer is rated 30MVA, 18KV/69KV, 8.5% impedance, and 27.3 X/R ratio. The step-down transformer is rated 5MVA, 69KV/4000V, 3.25% impedance, and 12.14 X/R ratio. Calculate the following quantities for this system.

- Real and reactive power supplied by the source. (2.330068MW, 1.035995MVAR)
- Load voltage. (3995.089V)
- Equivalent impedance of the load. ($6.353846\angle 22.628^\circ\Omega$)
- Voltage, current, and power factor on both ends of the transmission line. (69.7437KV, 36.450A, 0.916307 lagging, 69.4405KV, 36.450A, 0.917371 lagging)
- Real and reactive power flow on both ends of the transmission line. (2.329412MW, 1.018084 MVAR, 2.321981MW, 1.007473MVAR)
- Percent loading and efficiency of each transformer. (8.474%, 99.972% and 50.240%, 99.855%)
- Transmission line efficiency. (99.681%)
- Power system efficiency. (99.508%)

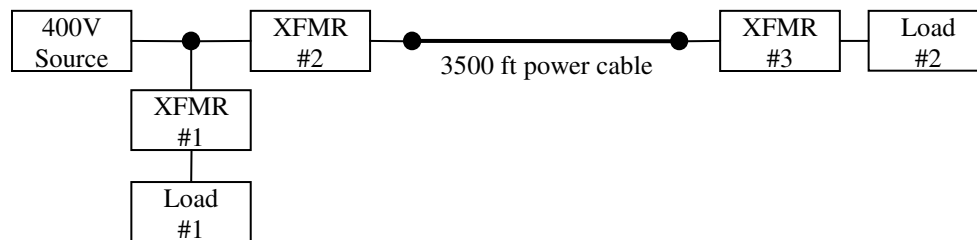
5. The one-line diagram for a three phase power system is shown below. In this system a generator supplies power to a remotely located load through a high voltage transmission line. One transformer is used to step-up the source voltage to create a high voltage level for the transmission line, and another transformer is used to step-down the high transmission line voltage to the level required for the load.



The generator operates at 23.215KV and supplies 8.465MVA at a 0.9995 leading power factor. The transmission line has a total per phase series impedance of $20\angle 85^\circ\Omega$. The step-up transformer is rated 15MVA, 23KV/69KV, Y- Δ , $r_1=0.08\Omega$, $x_1=0.9\Omega$, $r_2=0.4\Omega$, $x_2=5.6\Omega$. The step-down transformer is rated 10MVA, 69KV/4160V, Δ -Y, $r_1=0.5\Omega$, $x_1=6.3\Omega$, $r_2=0.01\Omega$, $x_2=0.07\Omega$. The load is Δ -connected with unknown per phase impedance. Compute the following quantities for this system.

- Real and reactive power supplied by the generator. (8.460768MW, -2.676533MVAR)
- Load voltage. (4187.346V)
- Per phase impedance of the load. ($6.23114\angle -6.972^\circ\Omega$)
- Voltage, current, and power factor on both ends of the transmission line. (69.5902KV, 70.174A, 0.998796 leading, 69.5398KV, 70.174A, 0.996473 leading)
- Real and reactive power flow on both ends of the transmission line. (8.448161MW, -4.148927MVAR, 8.422410MW, -7.092320MVAR)
- Percent loading and efficiency of each transformer. (18.796%, 99.851% and 28.139%, 99.488%)
- Transmission line efficiency. (99.695%)
- Power system efficiency. (99.037%)

- A block diagram for a single phase power system is shown below. In this power system, a source supplies power to two loads at two different locations and operating at two different nominal voltage levels.



The specifications for the power system components are given below.

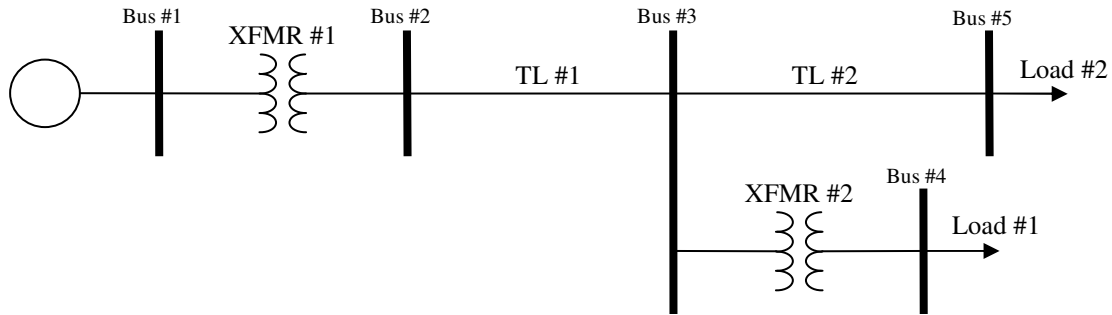
- Source:** Ideal, 400V
Load #1: Equivalent impedance of $5\angle 15^\circ\Omega$
Load #2: Equivalent impedance of $17\angle 75^\circ\Omega$
XFMR #1: 5KVA, 400V/120V with $r_1=0.9\Omega$, $x_1=1.6\Omega$, $r_2=0.002\Omega$, $x_2=0.004\Omega$
XFMR #2: 7.5KVA, 400V/1300V with 7.31% impedance and, 3.75 X/R
XFMR #3: 7.5KVA, 1300V/240V with $R_1=5.134\Omega$, $X_1=23.170\Omega$
Power Cable: 3500ft with distributed series impedance of $0.084\angle 61.5^\circ\Omega/\text{kft}$

Compute the following quantities for this system.

- Equivalent impedance of the system as seen by the source. ($30.8614\angle 47.475^\circ\Omega$)
- Real and reactive power supplied by the source. (3.5042KW, 3.8209KVAR)
- Voltage and current at each of the two loads. (117.189V, 23.438A and 221.989V, 13.058A)
- Voltage, current, and power factor on both ends of the power cable. (1.2603KV, 2.411A, 0.257028 lagging, 1.2596KV, 2.411A, 0.256900 lagging)
- Real power loss in the power cable. (0.815W)
- Percent loading of each transformer. (54.933%, 40.510%, 38.650%)
- Efficiency of each transformer. (98.310%, 96.938%, 96.175%)
- Power system efficiency. (97.120%)

- In the single phase power system shown in the one line diagram below, a 5KV source supplies power to two loads through a simple transmission line network. Transformer XFMR #1 is used to step-up the source

voltage to a higher level for transmission, and the other transformer XFMR #2 is used to step-down the voltage to a specific level required by one of the loads.

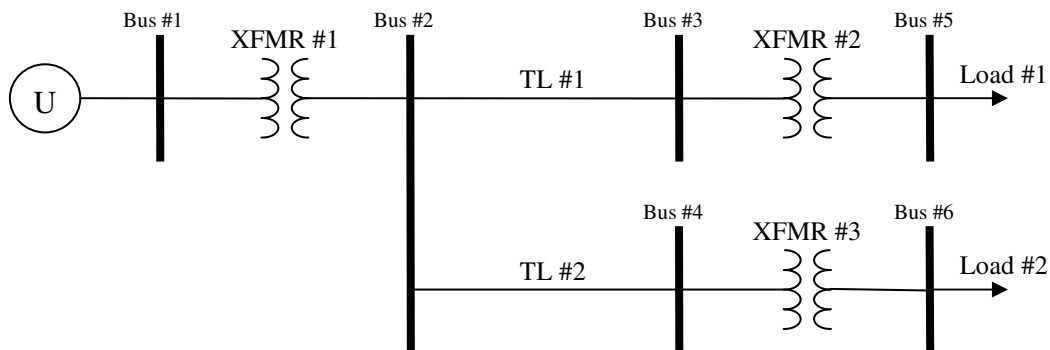


The total series impedances of the transmission lines TL #1 and TL #2 are respectively $0.9\angle 75^\circ\Omega$ and $1.7\angle 75^\circ\Omega$. The equivalent impedances of Load #1 and Load #2 are respectively $20\angle 15^\circ\Omega$ and $1950\angle 75^\circ\Omega$. The step-up transformer is rated 300KVA, 5KV/23KV, $r_1=0.9\Omega$, $x_1=1.6\Omega$, $r_2=5\Omega$, $x_2=12\Omega$, and step-down transformer is rated 3KVA, 23KV/240V, $r_1=5\Omega$, $x_1=12\Omega$, $r_2=0.05\Omega$, $x_2=1.9\Omega$.

Compute the following quantities for this system.

- Real and reactive power supplied by the source. (72.2950KW, 255.5762KVAR)
- Current in each of the two transmission lines. (11.548A, 11.480A)
- Voltage at each of the two loads. (226.741V, 22.3853KV)
- Percent loading and efficiency of each transformer. (86.284%, 95.565% and 85.686%, 99.739%)
- Power system efficiency. (95.433%)

8. The one line diagram and complete specifications are given below for a small three phase industrial power system in which a 485V tie from the electric utility company supplies power to two remotely located loads through high voltage power cables. Transformers are used to operate the power cables at high voltage and to create the desired voltage levels at the two loads.



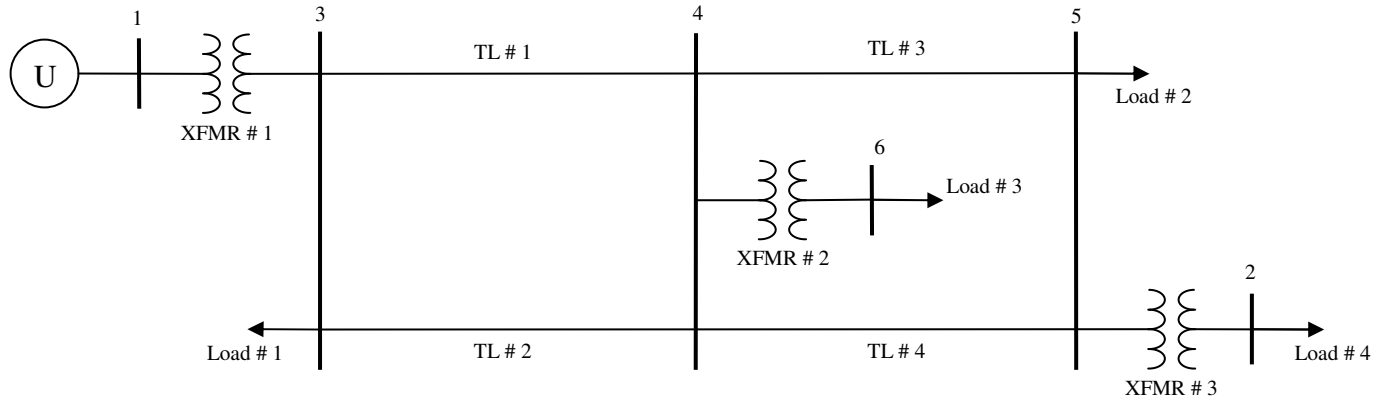
- Source:** 480V electric utility tie, source impedance $0.015\angle 85^\circ\Omega$, no-load voltage 486.25V
XFMR #1: 50KVA, 480V/4160V, Y-Y, 2.6% Z, 1.75 X/R
XFMR #2: 35KVA, 4160V/480V, Y- Δ , 2.6% Z, 1.69 X/R
XFMR #3: 20KVA, 4160V/208V, Y-Y, 2.3% Z, 1.13 X/R
TL #1: 3-conductor power cable with total per conductor impedance $1\angle 30^\circ\Omega$
TL #2: 3-conductor power cable with total per conductor impedance $1.5\angle 30^\circ\Omega$
Load #1: Δ -connected with per phase impedance $42\angle 75^\circ\Omega$
Load #2: Y-connected with per phase impedance $6\angle 35^\circ\Omega$

Compute the following quantities for this system.

- Voltage, current, and power factor at the electric utility company tie point. (485.606V, 26.743A, 0.455547 lagging)
- Real and reactive power supplied by the electric utility company to the system. (10.2467KW, 20.0238KVAR)

- c. Voltage and current at each of the two loads. (474.157V, 19.554A, 206.235V, 19.845A)
- d. Percent loading and efficiency of each of the three transformers. (44.473%, 98.755% and 45.883%, 97.651% and 35.444%, 99.334%)
- e. Power system efficiency. (97.233%)

9. The one line diagram and component specifications for a three phase power system are given below. In this system, an electric utility company supplies power to the four distributed loads of an industrial facility through a network of transmission lines and transformers.



- Source:** 231.5KV electric utility company tie with source impedance $0.995\angle 82^\circ \Omega$
- XFMR #1:** 50MVA, 230KV/34.5KV, $\Delta_{cba}-\Delta$, $r_1=0.5\Omega$, $x_1=0.1\Omega$, $r_2=0.01\Omega$, $x_2=0.1\Omega$
- XFMR #2:** 10MVA, 34.5KV/4160V, Y-Y, $r_1=0.01\Omega$, $x_1=0.1\Omega$, $r_2=0.009\Omega$, $x_2=0.08\Omega$
- XFMR #3:** 100KVA, 34.5KV/960V, Δ -Y, $r_1=0.01\Omega$, $x_1=0.1\Omega$, $r_2=0.008\Omega$, $x_2=0.02\Omega$
- TL #1:** Distributed per phase series impedance of $1\angle 80^\circ \Omega/\text{mile}$, 2.23 miles
- TL #2:** Same design as TL#1, 0.72 miles
- TL #3:** Same design as TL#1, 1.78 miles
- TL #4:** Same design as TL#1, 1.37 miles
- Load #1:** Δ -connected with per phase impedance $279\angle 75^\circ \Omega$
- Load #2:** Δ -connected with per phase impedance $555\angle 75^\circ \Omega$
- Load #3:** Y-connected with per phase impedance $10\angle 35^\circ \Omega$
- Load #4:** Y-connected with per phase impedance $12\angle 25^\circ \Omega$

Compute the following quantities for this system.

- a. Voltage at each of the six busses in the system. (231.4119KV, 956.675V, 34.6910KV, 34.5701KV, 34.4256KV, 4145.895V)
- b. Real and reactive power supplied by the electric utility company to the system. (6.498209MW, 19.786648MVAR)
- c. Current in each of the four transmission lines. (32.195A, 99.716A, 47.087A, 61.178A)
- d. Percent loading and efficiency of each of the three transformers. (41.628%, 99.961% and 17.188%, 99.888% and 76.269%, 99.926%)
- e. Power system efficiency. (99.787%)